POWER BI DATA MODELING & DAX SPECIFICATION

Executive Business Intelligence Implementation Guide | Petrochemical Trade Analytics

1. Star Schema Dimensional Architecture

Power BI Desktop connects to four pre-formatted relational tables located in data/powerbi/:
• **Fact_TradeData.csv:** Grain of one record per trade transaction with quantity, volume MT, value USD, and foreign key references.
• **Dim_Commodity.csv:** HS Codes, Chemical Groups (Olefins vs Aromatics), and descriptions.
• **Dim_Country.csv:** Reporter and partner economies with ISO codes and continental regions.
• **Dim_Date.csv:** Continuous time calendar from 2015 to 2024 with year, quarter, month, and month-year keys.

2. Essential DAX Measures Library

Measure Name	DAX Expression	Business Significance
Total Trade Value	SUM(Fact_TradeData[trade_value_usd])	Gross commercial market velocity ($302.18B)
Import Value	CALCULATE(SUM(Fact_TradeData[trade_value_usd]), Fact_TradeData[trade_flow] = "Import")	Total capital flowing into the port ($153.74B)
Export Value	CALCULATE(SUM(Fact_TradeData[trade_value_usd]), Fact_TradeData[trade_flow] = "Export")	Total capital flowing out of the port ($148.44B)
Trade Balance	[Export Value] - [Import Value]	Net trade surplus (+) or deficit (-) (-$5.29B)
Total Volume Mt	DIVIDE(SUM(Fact_TradeData[trade_volume_mt]), 1000000, 0)	Total physical shipment in Million MT (334.13 Mt)
Avg Price USD/MT	DIVIDE(SUM(Fact_TradeData[trade_value_usd]), SUM(Fact_TradeData[trade_volume_mt]), 0)	Trade value per metric ton
YoY Trade Growth	VAR Prev = CALCULATE([Total Trade Value], DATEADD(Dim_Date[full_date], -1, YEAR)) RETURN DIVIDE([Total Trade Value] - Prev, Prev, 0)	Year-over-year velocity

3. Step-by-Step 1-Click Power BI Desktop Setup Guide

1. Open **Power BI Desktop**.
2. Click **Get Data -> Text/CSV** and select all 4 files from data/powerbi/.
3. In the **Model View**, verify the 1-to-Many relationships from Dim tables to Fact_TradeData.
4. In the **Report View**, create a New Table named _Measures and paste the DAX formulas above.
5. Drag and drop the **Card Visual** for Total Trade, Imports, Exports, and Trade Balance.
6. Add a **Clustered Column Chart** for Annual Trade Value by Year and Trade Flow.